LUKMANUL HAKIM

Principal Solutions Architect & Cloud Modernization Strategist
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PROFESSIONAL SUMMARY

Cross-layer Solutions Architect with a career spanning telecom infrastructure (since 2011) and multi-cloud platform engineering. Specializes in legacy modernization, monolith decomposition, and integrating generative AI and agentic systems into resilient, cloud-native microservices. Bridges physical failure modes (L1 fiber/FTTH) to distributed platforms (L7 Kubernetes/GitOps) using IEC 60812 (FMEA), IEC 61025 (FTA), and ISO 31000 to eliminate single points of failure (SPOF). Empaneled expert with VisasQ and Coleman Research, delivering technical due diligence and modernization assessments for private equity and institutional investors.

CORE SKILLS

Cloud & Platforms: Google Cloud Platform (GCP), Microsoft Azure, Amazon Web Services (AWS), Alibaba Cloud, Multi-Cloud, Hybrid Cloud, High Availability (HA), Disaster Recovery (DR), Workload Migration
Containers & Automation: Kubernetes (GKE/EKS), Docker, Helm, Terraform, Ansible, GitOps (Flux CD), CI/CD (Jenkins, GitHub), Infrastructure as Code (IaC), Zero-Touch Provisioning (ZTP)
Architecture & Data: Microservices, Event-Driven Architecture, Pub/Sub, Legacy Modernization, Monolith Decomposition, BigQuery, PostgreSQL, MySQL, Oracle, ETL Data Reconciliation
AI & Agentic Systems: Azure AI Studio, Azure OpenAI, Semantic Kernel, AI Agents, LLM Orchestration, Prompt Engineering, Prompt Defense & Safety, AI Ethics, NLP, Computer Vision, TensorFlow
Reliability & Risk: IEC 60812 (FMEA), IEC 61025 (FTA), ISO 31000, SPOF Elimination, Root Cause Analysis (RCA), MTTR/MTBF Optimization, Site Reliability Engineering (SRE), Technical Due Diligence, FinOps
Security & Networking: IAM, RBAC, Zero Trust, DevSecOps, SonarQube, OWASP ZAP, Secret Management, OJK & Bank Indonesia Compliance, MPLS, MikroTik, SolarWinds, ELK Stack
Languages & Systems: Python, Go, Bash, PowerShell, YAML, JSON, Linux/RHEL, Windows Server

PROFESSIONAL EXPERIENCE

Independent Expert Advisor (Empaneled) **Sep 2025 – Present**
VisasQ Inc. & Coleman Research Group (Expert Networks) | Global (Remote)

- Deliver retained, NDA-governed technical due diligence and architecture modernization assessments for private equity firms and institutional investors evaluating cloud migration feasibility and telecom infrastructure acquisitions.
- Build quantitative risk exposure models, fault isolation strategies, and SPOF mitigation frameworks for legacy-to-cloud transitions using IEC 60812 (FMEA) and IEC 61025 (FTA).
- Advise executive leadership on multi-cloud cost governance (FinOps), scalability, and technical debt remediation ahead of major capital deployments.
- Assess AWS, GCP, and Azure platforms, microservices resilience, and data infrastructure integrity to protect operational reliability and EBITDA.

Key Competencies & Technologies: Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure, Multi-Cloud, Cloud Migration, Technical Due Diligence, IEC 60812 (FMEA), IEC 61025 (FTA), ISO 31000, FinOps, Scalability, Microservices Resilience, Data Governance, Telecom Infrastructure

Enterprise IT Advisory Consultant **Jan 2023 – Feb 2025**
PT Altha Dyanusa Consulting | Jakarta, Indonesia

- Designed enterprise multi-cloud blueprints and ISO 31000-aligned IT risk governance frameworks for clients moving mission-critical monolithic core systems to cloud-native architectures.
- Advised engineering leadership on zero-downtime migration pathways, container security compliance policies, and fault-tolerant cloud economics.
- Standardized DevSecOps policies across hybrid environments, with automated compliance controls and baseline vulnerability SLAs.

Key Competencies & Technologies: Cloud Architecture, Multi-Cloud Strategy, ISO 31000, DevSecOps Governance, Container Security, Zero-Downtime Migration, Infrastructure Optimization, Compliance Standards

Cloud Migration Lead & Solutions Consultant **Sep 2022 – Dec 2022**
PT Meteor Inovasi Digital | Jakarta, Indonesia

- Led a 4-month multi-cloud legacy modernization across AWS, GCP, and Alibaba Cloud for high-traffic enterprise applications.
- Decomposed VM-based monolithic workloads into containerized Kubernetes microservices with Terraform, achieving 99.9%+ availability and a zero-downtime cutover.
- Built automated deployment pipelines and multi-region failover, reducing single-cloud dependency risk.

Key Competencies & Technologies: Kubernetes, Terraform, Infrastructure as Code (IaC), Multi-Cloud Migration, Disaster Recovery, Microservices, Alibaba Cloud, GCP, AWS

FinTech Solutions & DevSecOps Architecture Lead **Jul 2022 – Sep 2022**

PT Kredit Pintar Indonesia | Jakarta, Indonesia

- Architected GitOps continuous delivery with Flux CD on Alibaba Cloud, decoupling stateful legacy FinTech backends into scalable stateless microservices.
- Enforced automated audit verification pipelines and container hardening aligned with OJK and Bank Indonesia data governance standards.
- Implemented RBAC and zero-trust telemetry across financial transaction pathways.

Key Competencies & Technologies: GitOps (Flux CD), Alibaba Cloud, DevSecOps, FinTech Systems Architecture, Microservices Decomposition, RBAC, Zero Trust Telemetry, OJK & Bank Indonesia Compliance

Container Platform & DevSecOps Consultant **Nov 2021 – Jul 2022**

PT Tri Adi Bersama (Anteraja) | Jakarta, Indonesia

- Scaled multi-cluster Kubernetes (GKE/EKS) across 120 logistics microservices handling peak nationwide delivery traffic.
- Integrated CI/CD security gates (OWASP ZAP, SonarQube), eliminating 60% of critical vulnerabilities before production.
- Tuned pod autoscaling and compute allocation, reducing infrastructure overhead while sustaining sub-second API latency.

Key Competencies & Technologies: Kubernetes (GKE/EKS), Container Orchestration, CI/CD Security, OWASP ZAP, SonarQube, Vulnerability Scanning, Pod Autoscaling, Performance Optimization, Logistics Microservices

Cloud Automation & Infrastructure Lead **May 2020 – Oct 2020**

PT Hashmicro Solusi Indonesia | Jakarta, Indonesia

- Built declarative Terraform provisioning workflows for multi-tenant enterprise ERP cloud deployments.
- Cut client environment provisioning lead time by 70% through modular zero-touch provisioning automation.

Key Competencies & Technologies: Infrastructure as Code (IaC), HashiCorp Terraform, Cloud Automation, Zero-Touch Provisioning (ZTP), ERP Architecture, Multi-Tenant Deployments

DevOps Infrastructure Consultant **Jun 2019 – Sep 2019**

PT KB Finansia Multi Finance | Jakarta, Indonesia

- Optimized hybrid cloud and on-premise VPS infrastructure for high-availability financial services platforms.
- Accelerated CI/CD release cycles by 40% with containerized Jenkins pipelines integrated with Jira and Docker.
- Improved observability with a containerized ELK Stack, enabling faster incident resolution and lower MTTR.

Key Competencies & Technologies: Hybrid Cloud, Continuous Integration (CI), Release Management, Jenkins, Docker, Observability & Telemetry, ELK Stack, MTTR Optimization

Network Telemetry & Infrastructure Engineer **Feb 2015 – Jun 2015**

PT Dimension Data Indonesia | Jakarta, Indonesia

- Optimized MPLS core backbone performance for Tier-1 banking clients using SolarWinds telemetry, removing latency bottlenecks and transient packet loss in payment clearing and interbank settlement systems.

Key Competencies & Technologies: Network Performance Optimization, Enterprise Network Telemetry, SolarWinds, MPLS Core Backbone, Financial Payment Infrastructure, Transmission Engineering

Principal Operational and Compliance Manager **Apr 2013 – Oct 2014**

PT Telkom Akses (Telkom Indonesia Group) | Jakarta, Indonesia

- Executed field quality compliance audits and service fulfillment testing for a nationwide copper-to-FTTH rollout (IDR 1.5T program scope, approx. $93M at current exchange rate).
- Applied IEC 61025 (FTA), IEC 60812 (FMEA), and ISO 31000 to audit fulfillment failure modes, driving a 20% defect reduction during integration testing with zero capital asset discrepancy.
- Developed automated ETL reconciliation scripts (Oracle, PostgreSQL, SOAP APIs) linking field infrastructure inventory with BSS/OSS platforms (I-Sisca, TenOSS) to keep asset registers audit-compliant.

Key Competencies & Technologies: Application Lifecycle Management (ALM), Asset Management, Operational Compliance, Nationwide FTTH Rollout, IEC 61025 (FTA), IEC 60812 (FMEA), ISO 31000 Risk Governance, ETL Reconciliation, Oracle, PostgreSQL, SOAP APIs

Electronic Data Processing Specialist **Jan 2013 – Mar 2013**

Circus Waterpark Kuta | Bali, Indonesia

- Secured POS and ticketing networks with MikroTik firewalls, and automated POS data synchronization and ticketing gateway processing with Bash and PowerShell.
- Administered Windows Server environments and CCTV telemetry; coordinated system security work within an IDR 200M infrastructure upgrade.

Key Competencies & Technologies: MikroTik Firewalls, Perimeter IT Security, Bash & PowerShell Automation, Windows Server Administration, POS & Ticketing Systems, CCTV Telemetry, Access Control

Customer Care Specialist **Oct 2011 – Oct 2012**

Biznet Networks | Jakarta, Indonesia

- Resolved enterprise L1/L2 network and fiber incidents and SLA escalations; coordinated cross-functional root cause analysis (RCA) and ticket resolution through enterprise CRM.

Key Competencies & Technologies: Escalation Management, Incident Resolution, Network Support (L1/L2), Fiber Infrastructure SLAs, Root Cause Analysis (RCA), Enterprise CRM

CERTIFICATIONS

Microsoft Certified: Azure AI Apps and Agents Developer Associate (AI-103) | Credential ID: MS1100623380
Google Cloud Certified: Professional Cloud Architect | Credential ID: ac6c40d564fc4911b48506b04f269dac
Google Cloud Certified: Associate Cloud Engineer | Credential ID: 55864fb403294d648d9d213032600842
AWS Certified: Cloud Practitioner | Credential ID: YT3X1MX1VFBE1K5V
Cisco Certified: DevNet Associate | Verification: 1987298840-45/PROA/BLSDM/2024
Cisco Certified: CyberOps Associate | Verification: 0447256131-103/OA.DTS/2020

TRAINING

Microsoft Azure AI & Data: AI-102 (Azure AI Solutions, Generative AI Knowledge Mining), DP-100 (Data Science, Azure OpenAI, Semantic Kernel)
Microsoft Applied AI: AI-3002 (Document Intelligence), AI-3003 (Natural Language Processing), AI-3004 (Computer Vision)
AWS Cloud Quest: Generative AI Practitioner (learning badge) | Credential ID: 319f5386-8769-4719-693f-29b81814900c
Azure DevOps Pipeline Security; Serverless Container Architecture

EDUCATION

Bachelor of Economics (S.E.), Universitas Mercu Buana **Graduated Oct 2018**
Jakarta, Indonesia | GPA 3.17 / 4.00
Thesis: Operational Risk Analysis Using FTA and FMEA in the Process Testing of the FTTH Triple Play (IndiHome) Fulfillment System